

SAHIL KUMAR

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Education

Indian Institute of Information Technology (IIIT)

Aug. 2024 – Aug. 2028

Bachelor of Technology in Computer Science

Jabalpur, MP

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Computer Networks
- Operating Systems
- Database Management
- Computer Architecture
- Data Science
- Embedded Systems

Technical Skills

Languages: C++, Python, Java, SQL, C, JavaScript, HTML/CSS

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Computer Networks, Operating Systems

ML/Data Tooling: NumPy, Pandas, Scikit-learn, FAISS, LangChain, DVC, MLflow, RAG architectures, multi-stage pipeline design

Developer Tools: Git, GitHub, Linux, VS Code, Jupyter Notebooks

Projects

NEWS_CLASSIFIER – End-to-End MLOps Pipeline

Project

Python, Scikit-learn, DVC, MLflow, TF-IDF

Link

- Designed and implemented a complex, multi-stage ML system (data versioning, training, evaluation, inference) on the AG News dataset, achieving **91.77% test accuracy** via TF-IDF feature extraction and GridSearchCV-tuned classifiers.
- Built a fully reproducible, parameterized pipeline (`params.yaml`, `train.py`, `predict.py`, `dvc.yaml`) with DVC for data/model versioning and MLflow for experiment tracking and comparison across runs.
- Engineered the system as discrete, independently testable stages to support reliability and reproducibility, mirroring real-world ML infrastructure design rather than a single monolithic script.

NyayAI – Retrieval-Augmented Legal Document System

Team Project

Python, LangChain, FAISS

Link

- Building a RAG pipeline that converts unstructured Indian legal documents into clean, embedded, retrievable chunks for downstream reasoning systems, using LangChain for orchestration and FAISS for vector similarity search.
- Collaborating in a multi-contributor open-source workflow (branching, PR review, issue tracking), focusing on the retrieval and preprocessing stages of the pipeline.

Whoosh – Peer-to-Peer Encrypted File & Text Transfer

In Progress

WebRTC, DTLS, AES-GCM

- Designing a P2P transfer system using WebRTC data channels with DTLS/AES-GCM encryption, STUN/TURN NAT traversal, and concurrent chunked transfer for large files; targeting deployment Aug 2026.

Experience

Ambao Krishi Kutumb

Jan. 2025 – Feb. 2025

Internet of Things (IoT) Intern

Remote

- Designed IoT workflow architecture for sensor data collection and monitoring.
- Assisted in deployment planning and field testing of IoT hardware systems.

The Programming Club (TPC)

Aug. 2024 – Present

Active Member / Competitive Programmer

IIIT Jabalpur

- Solved 600+ data structures and algorithms problems across Codeforces and CodeChef, with consistent focus on debugging and optimizing runtime/space performance in C++.
- Participate in weekly internal contests; collaborate with peers on problems involving Number Theory, Combinatorics, Graphs, and Dynamic Programming.
- Mentored and organized beginner-level coding sessions and workshops for the 2025 incoming batch.

Extra-curricular

Programming Mentor: Guiding a group of motivated freshers into competitive programming and software development.

Samvaad: Member of the literary society, organizing and participating in college-wide events.

Achievements / Profiles

Codeforces: Rating: Pupil (Max 1239) – Consistently competing in Div. 2 and Div. 3 contests.

CodeChef: 3-Star (Max 1608) – Achieved ranks under 500 in multiple Division 3 contests.

Problem Solving: 600+ problems solved across platforms, focused on Arrays, Strings, Greedy, DP, and Graphs.